巴适声工厂隐私版使用手册 / Bashi Voice Factory Privacy Edition User Guide

Bashi Voice Factory Privacy Edition (巴适声工厂 · 隐私版)

License MIT version 0.1.3 python 3.12 embed platform Windows

Version: 0.1.3

A fully offline desktop web app for high-quality text-to-speech and speech-to-text. After the one-time first-launch download, **everything runs on your own machine** — TTS synthesis, audio export, transcription, and storage. No audio data ever leaves your computer.

> ⚡ Need cloud-quality voices and faster generation? Check out Bashi Voice Factory Turbo (巴适声工厂 · 极速版) — Microsoft Edge TTS, 14 languages, 50,000-character long text. Requires internet.

Author: Alex Li (ncorecpu@gmail.com) **License:** MIT License **Source code:** https://github.com/gtree965/bashi-voice-factory-privacy **Download:** GitHub Releases · files.fm mirror

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🆕 What's New in v0.1.3

- **Safer STT defaults:** SenseVoice Small is now the default multilingual STT model, the Speaker ID UI is disabled, and Paraformer Chinese Large has been removed from the product.

- **Upload and job-state hardening:** a server-side upload ceiling (2 GB by default) rejects oversized requests, and shared `stt_jobs` state is now lock-protected across background work and API reads.
- **Reliable streaming:** affected frontend SSE readers now buffer data across network chunks, preventing split JSON lines or frames from being dropped or misparsed.
- **Responsive streaming synthesis:** stop long or chunked synthesis at any time without losing completed chunks; they remain playable and individually downloadable. In Shadowing Mode, playback starts when the first chunk is ready by default (this can be disabled) and waits for later chunks instead of stopping early.
- **Listen while long text is generated:** GGUF long-form synthesis now previews the first ready audio group and continues seamlessly as later groups arrive, with mute and adaptive buffering controls. The complete seekable/downloadable MP3 still appears when merging finishes, without interrupting an active preview.

✨ Highlight Features

🔒 Fully Local Text-to-Speech

- **Local Qwen3-TTS-12Hz-1.7B-CustomVoice** running entirely on your machine — no cloud calls during synthesis.
- **9 curated voice presets** with native-language sample previews built in.
- **10 languages** supported: Chinese (Mandarin, Cantonese, Sichuanese, Beijing, Northeast), English (US/UK), Japanese, Korean, German, French, Russian, Portuguese, Spanish, Italian.
- **Instruction control** for style, accent, emotion via natural-language prompts.
- **Streaming controls for long text and Shadowing Mode**: stop synthesis while keeping completed chunks, auto-play the first ready chunk by default, and continue playback as later chunks arrive; GGUF long text also preserves a complete seekable/downloadable MP3 after the streamed preview.

🎯 Automatic Backend Selection

- **GGUF + Vulkan** for AMD / Intel / iGPU users (default, fast, RAM-friendly).

- **GGUF + CUDA** for NVIDIA users via a one-click in-app upgrade (optional ~595 MB add-on download; no manual weight setup). New in v0.1.1.
- **Reliable NVIDIA detection and startup probes**: v0.1.2 detects the physical NVIDIA GPU through virtual-display layers and isolates native probe crashes so the launcher can report an actionable error instead of disappearing.
- **CPU fallback** for entry-level hardware (works on Intel N100-class boxes).
- One-click **speed test** calibrates ETA estimates to your specific machine.

🎙️ Local Offline Speech-to-Text (STT)

- **SenseVoice Small** (default, 242 MB) — multilingual zh/en/ja/ko/yue, non-autoregressive (fast on CPU).
- **Parakeet TDT 0.6B** (optional, 661 MB) — English specialist, NVIDIA's ~1.7% WER champion.
- **VAD-based segmentation** via Silero — accurate timestamps, no chunk-boundary stutter.
- **Live transcript streaming** via SSE; exports to TXT / SRT / VTT.

🛡️ Smart First-Launch Download with Resume

- **HTTP Range / resume** built in — a transient WiFi drop won't restart your 2.2 GB GGUF download from zero.
- **3-attempt retry** with exponential backoff for both pip dependencies and model files.
- **China-friendly mirrors**: pip falls back through Aliyun → Tsinghua → PyPI when a mirror is unreachable or refuses your IP; GGUF runtime hosted on ModelScope; STT models hosted on hf-mirror.cn.

📱 LAN Sharing for Phone / Tablet Use

- First-launch prompt: bind to `0.0.0.0` to allow access from any device on the same WiFi.
- Compatible with mobile browsers — type the LAN IP and use it like a local web app.

🚀 Quick Start

Easiest path

1. Unzip the downloaded file to any folder (Desktop or wherever you like).
2. **Double-click `start_启动.bat`** at the top level of the extracted folder.
3. First launch will:
   - Install Python dependencies (~700 MB, 8-15 min depending on network/CPU)
   - Automatically download the GGUF model when missing (~2.2 GB, 2-15 min depending on network; Ctrl+C cancels and the next launch resumes)
   - Open `http://127.0.0.1:5050` in your default browser
4. After first launch, **all subsequent launches are offline** and start in ~5 seconds.

Advanced

Users who prefer to bypass the top-level launcher can run `bashi-privacy-app\run_portable.bat` directly. Both paths are identical.

🌐 Network Behavior

Privacy commitment: this app does not phone home, does not upload audio, does not collect telemetry. **The app never automatically checks any internet resource.** Every network operation is documented below and requires explicit user action — no background polling, no analytics, no silent update probes.

First launch only (one-time, ~700 MB + ~2.2 GB):

- pip dependencies from `https://mirrors.aliyun.com/pypi/simple/`, falling back to Tsinghua then PyPI (PyPI only, outside China)
- GGUF model from `https://modelscope.cn/models/gtree592/bashi-qwen3-tts-1.7b-customvoice-gguf-runtime`
- STT model (when user opts in) from `https://hf-mirror.com/csukuangfj/...` (or HuggingFace fallback)

Runtime network activity: zero. TTS synthesis, STT transcription, and audio export are 100% on-device.

Optional CUDA acceleration (NVIDIA users, user-initiated only): when an NVIDIA GPU is detected, the UI shows an "Upgrade to CUDA acceleration" banner. Clicking it downloads a one-time ~595 MB CUDA runtime add-on from `https://modelscope.cn/models/gtree592/bashi-qwen3-tts-cuda-runtime`. This is entirely optional — the Vulkan path works on NVIDIA without it. Nothing downloads unless you click.

Manual update check (user-initiated only, never automatic): the UI footer "Check for updates" button opens a release page in a new browser tab. You can also check manually anytime at:

- GitHub Releases: https://github.com/gtree965/bashi-voice-factory-privacy/releases
- files.fm mirror: https://files.fm/u/juvstxmrez

Air-gapped / sensitive deployment (banks, healthcare, government, internal corporate networks): after first-launch downloads complete on a connected machine, the entire `bashi-voice-factory-privacy-v0.x.0/` extracted folder can be copied to an offline / physically-isolated machine and run with zero further network access required. The app has no telemetry endpoints to firewall, no auto-update worker, no callback URLs. To verify, monitor outbound traffic with Wireshark / Resource Monitor during a normal session — it should be silent.

💻 Hardware Expectations

The table below shows **actually measured** numbers from author hardware. The built-in speed test (click 测速 / Speed Test on the right-hand panel) calibrates the ETA panel to your specific machine on first use, so you don't have to extrapolate from this table.

Hardware	Auto-selected backend	"你好。" probe (25 chars)	1,000-char synthesis ETA	Status
AMD RX 9060 XT (16 GB)	GGUF + Vulkan	~3 s	a few minutes	✓ Tested 2026-05
AMD RX 590 (8 GB)	GGUF + Vulkan	3-5 s	~5-10 min	✓ Tested

Hardware	Auto-selected backend	“你好。” probe (25 chars)	1,000-char synthesis ETA	Status
Intel N305 laptop + UHD iGPU	GGUF + Vulkan / DirectML	53 s	25-46 min	✓ Tested 2026-05-25
Intel N100 mini-PC + UHD iGPU	GGUF + Vulkan / DirectML	126 s	58 min - 1h49 min	✓ Tested 2026-05-25
NVIDIA RTX 5070 (12 GB, cloud PC, Blackwell)	GGUF + Vulkan/CUDA + DirectML	~1 s	41 s - 1 m 17 s	✓ Tested 2026-06-12 (Vulkan) & 2026-07-24 (海马云 HMv Cloud PC): full cold-start synthesis confirmed; optional CUDA add-on verified working (LLM 28 layers on CUDA0, decoder DirectML fp16). CUDA-vs-Vulkan A/B still welcome.
Other NVIDIA RTX / GTX (desktop)	GGUF + Vulkan + DirectML · optional CUDA add-on	community reports welcome	community reports welcome	awaiting desktop tester reports via GitHub Issues
Apple Silicon / Intel Arc	—	not yet measured	not yet measured	not validated yet

> ⚠️ **Entry-level CPUs (Intel N100 / N305 class) are only practical for short text — under ~200 characters per generation.** A 5,000-character essay would take 2-9 hours on these boxes. For long-form audio (lectures, audiobooks), please use discrete-GPU hardware (AMD RX 500/600/9000 series, NVIDIA RTX class).

> ℹ️ **NVIDIA users (desktop RTX / GTX)**: the default path is GGUF + Vulkan (NVIDIA cards support Vulkan via the proprietary driver). For native CUDA acceleration, click the one-click in-app upgrade banner (v0.1.1+) — it downloads a ~595 MB CUDA runtime add-on, no manual weight setup required. **Requires NVIDIA driver ⩾ 545.x for the CUDA 12.4 runtime.** Vulkan path measured on RTX 5070 (cloud PC): **1 s for the 25-char probe**, so the default Vulkan experience is already excellent on a desktop NVIDIA card. CUDA add-on A/B numbers + reports from other RTX/GTX cards welcome via GitHub Issues. Cloud datacenter NVIDIA cards (A10/A100/T4) running in TCC mode have additional setup steps — see Troubleshooting below.

🧩 Hardware Coverage Matrix

Beyond the specific machines benchmarked above, here is how the auto-selected acceleration path maps across mainstream Windows hardware:

Hardware class	Acceleration path used	Status
NVIDIA RTX 30 / 40 / 50 (desktop)	GGUF + Vulkan default • optional CUDA in-app upgrade	✅ RTX 5070 (Blackwell) verified on 海马云 cloud PC — 2026-06-12 Vulkan probe ~1 s; 2026-07-24 full cold-start synthesis + optional CUDA add-on confirmed working (LLM 28 layers on CUDA0, decoder DirectML fp16, driver 610.47). CUDA-vs-Vulkan A/B numbers + other-card reports still welcome via Issues.

Hardware class	Acceleration path used	Status
		CUDA add-on requires driver ⩾ 545.x.
NVIDIA GTX 10 / 16 (desktop)	GGUF + Vulkan default • optional CUDA in-app upgrade	✅ Same flow, same driver requirement; reports welcome via Issues.
NVIDIA datacenter (A10 / A100 / T4)	Manual setup required	⚠️ TCC mode + old cloud driver = manual workaround. See Troubleshooting.
AMD RX 500 / 600 / 7000 / 9000 (discrete)	GGUF + Vulkan + DirectML	✅ Tested (RX 590, RX 9060 XT)
Intel Arc A-series (A380 / A580 / A750 / A770)	GGUF + Vulkan + DirectML	✅ Should work — not yet measured
Intel iGPU (UHD / Iris Xe / Arc iGPU)	GGUF + Vulkan + DirectML	✅ Tested (Intel N305 UHD)
AMD APU iGPU (Vega 7/8, RDNA 2/3)	GGUF + Vulkan + DirectML	✅ Should work — not yet measured
CPU only (no usable GPU / no driver)	GGUF + CPU (ggml-cpu auto-SIMD)	✅ Works; auto-selected only when no GPU detected
NPU (Snapdragon X / Lunar Lake / Ryzen AI 300)	—	❌ Not yet utilized
ARM64 Windows (Snapdragon X Copilot+ PC)	—	❌ Not supported (runs via slow x64 emulation)

⚙️ Known Limitations (roadmap)

- **NVIDIA CUDA is opt-in, not bundled.** By default NVIDIA cards use the Vulkan path; native CUDA acceleration requires the one-click in-app upgrade (~595 MB add-on, v0.1.1+). This keeps the main download small for the AMD / Intel / CPU majority. Requires NVIDIA driver ⩾ 545.x.
- **Weak iGPU may be slower than pure CPU.** On Intel N100-class hardware, DirectML / Vulkan transfer + scheduling overhead can outweigh

GPU benefit. A/B test by launching from a cmd window after `set GGUF_LLM_USE_GPU=0 && set GGUF_ONNX_PROVIDER=CPU` — feedback on real speed numbers welcome.

- **NPU acceleration is not yet used** (Snapdragon X Elite, Intel Lunar Lake AI Boost, AMD Ryzen AI 300). Under investigation for v0.2+.
- **ARM64 Windows is not natively supported** (Surface Pro 11, Galaxy Book4 Edge, etc.). x64 emulation works but is slow. Native ARM64 build is a v0.3+ candidate.

🐛 Troubleshooting

Symptom	Likely cause / fix
`Start_启动.bat` shows "Array index expression is missing"	Old zip — re-download the latest from GitHub Releases c files.fm mirror; the BOM bug was fixed in v0.1.0 final
pip install stops mid-way after WiFi blip	Retry triggers automatically (3 attempts, 5s/30s/120s backoff). If all 3 fail, fix network and re-run the launcher pip caches what's already installed
pip install fails with a "long path support" message	Run the registry one-liner from the launcher's advisory an Administrator PowerShell, then re-launch
pip install fails on every mirror, or a mirror refuses your IP (`403` / `denied by IP ACL`)	The launcher already retries Aliyun → Tsinghua → PyPI. force one index, set `BASHI_PIP_INDEX_URL` before launching e.g. `set BASHI_PIP_INDEX_URL=https://pypi.tuna.tsinghua.edu.cn/si`
GGUF download interrupted	Same: retries with HTTP Range/resume; re-run launcher continue from `.part` file
App exits with "No usable backend was found"	Check `app.log` (structured, timestamped application logs `launch_log.txt` (launcher steps and raw stderr) — usually GGUF runtime DLL missing, GPU driver outdated, or RAM GB. App now prints a bilingual advisory with specific cau
STT download says "镜像失败"	Should not happen in v0.1.0 final — old behavior from removed ModelScope path

Symptom	Likely cause / fix
Cloud / datacenter NVIDIA GPU (A10 / A100 / T4 on Chinese cloud Windows images): `access violation reading 0x0000000000000000` or `GGUF probe failed`	Datacenter NVIDIA GPUs typically run in **TCC mode** (Vulkan/DirectML disabled) with older drivers (~538.x) incompatible with the CUDA 12.4 add-on. **Desktop RTX/(cards are NOT affected** (they run WDDM mode by defaul with modern drivers). In v0.1.2, native startup-probe cras are isolated and reported instead of terminating the launcher, but TCC setup still requires the manual workaround: (1) rename `vulkan_backend_spike\Qwen3-TTS-GGUF\qwen3_tts_gguf\inference\bin\ggml-vulkan.dll` to `.disabled`; (2) edit `bashi-privacy-app\run_portable.ps1` and `$env:USE_GGUF_BACKEND = "1"` + `$env:GGUF_ONNX_PROVIDER = "` after the `Remove-Item Env:USE_GGUF_BACKEND` lines (~line 27(pre-install the CUDA add-on via CLI `python download_cuda_runtime.py`. CUDA 12.4 requires NVIDIA driv 545.x. Automatic TCC handling remains planned for a lat patch.

Full log paths: `bashi-privacy-app\app.log` and `bashi-privacy-app\launch_log.txt`

📦 What’ s in the Zip

```
bashi-voice-factory-privacy-v0.1.3/
├── Start_启动.bat                                        ← double-click here
├── Start_CPU_only_仅CPU启动.bat                          ← force CPU mode
(entry-level iGPU A/B)
├── README.md                                            ← this file
├── README_CN.md                                         ← Chinese version
├── LICENSE                                              ← MIT license
├── VERSION                                              ← release version
├── 巴适声工厂隐私版使用手册
_Bashi_Voice_Factory_Privacy_Edition_User_Guide.pdf
│                                                         ← help PDF
(bilingual)
├── bashi-privacy-app/                                   ← app code +
embedded Python
│   ├── run_portable.bat                                 ← same launcher,
direct path
│   └── ...
└── vulkan_backend_spike/                                ← GGUF runtime
(populated on first launch)
    └── Qwen3-TTS-GGUF/
```

📄 License

Bundled third-party components retain their original licenses:

- **Qwen3-TTS-12Hz-1.7B-CustomVoice**: Tongyi Lab (Alibaba), Apache 2.0
- **GGUF runtime**: based on HaujetZhao/Qwen3-TTS-GGUF
- **SenseVoice Small / Silero VAD / Parakeet TDT**: see respective model cards

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👤 Author

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Issues, feedback, and feature requests welcome via the GitHub release page or email.

巴适声工厂 · 隐私版 (Bashi Voice Factory Privacy Edition)

License MIT version 0.1.3 python 3.12 embed platform Windows

版本： 0.1.3

完全离线运行的本地语音工厂网页应用。首次启动联网下载完依赖与模型之后，**所有文字转语音、语音转文字、音频生成、文件保存都在你自己的电脑上完成**，没有任何音频数据上传云端。

> ⚡ 想要云端语音质量、更快的合成速度？请关注 巴适声工厂 · 极速版 (Bashi Voice Factory Turbo) — 微软 Edge TTS 引擎、14 种语言、5 万字长文。需要联网。

作者： Alex Li (ncorecpu@gmail.com) **许可：** MIT License **源码：** https://github.com/gtree965/bashi-voice-factory-privacy **下载：** GitHub Releases · files.fm 镜像

🆕 v0.1.3 更新内容

- **更稳妥的 STT 默认方案：** SenseVoice Small 现为默认多语种 STT 模型，Speaker ID 界面已停用，并从产品中移除了 Paraformer 中文大模型。
- **上传与任务状态加固：** 新增服务端上传硬上限（默认 2 GB），并为共享 `stt_jobs` 状态加入线程锁，覆盖后台任务与 API 读取路径。
- **更可靠的流式处理：** 受影响的前端 SSE 读取器现在会跨网络 chunk 缓冲数据，避免被拆开的 JSON 行或事件帧丢失、误解析。
- **更顺手的流式合成：** 长文或跟读片段合成可随时停止，已生成片段不会丢失，仍可逐段播放和下载。跟读模式默认在首个片段生成后自动开始播放（可关闭），播放追上合成时会等待后续片段，不再提前停住。
- **长文边合成边试听：** GGUF 长文现在会在首组音频就绪后开始试听，并随后续组自动接续；支持静音试听与慢机自适应缓冲。合并完成后仍提供可拖动、可下载的完整 MP3，且不会打断正在进行的试听。

✨ 核心亮点

🔒 完全本地文字转语音

- **Qwen3-TTS-12Hz-1.7B-CustomVoice 本地推理** — 合成过程零云端调用。
- **9 个精选音色**，内置母语试听样例。
- **10 种语言**：中文（普通话、粤语、四川话、北京话、东北话）、英文（美/英）、日语、韩语、德语、法语、俄语、葡萄牙语、西班牙语、意大利语。
- **自然语言指令控制** — 风格、口音、情绪都可通过 instruct 提示调整。
- **长文与跟读流式控制** — 可停止合成并保留已完成片段；默认自动播放首个就绪片段，后续片段生成后自动续播；GGUF 长文试听结束后仍保留可拖动、可下载的完整 MP3。

🎯 后端自动选择

- **GGUF + Vulkan**：AMD / Intel / 集成显卡用户默认路线（速度快、内存友好）。
- **GGUF + CUDA**：NVIDIA 用户一键应用内升级（可选 ~595 MB 附加包下载，无需手动配置权重）。v0.1.1 新增。
- **更可靠的 NVIDIA 检测与启动探测**：v0.1.2 可穿透虚拟显示层识别真实 NVIDIA 显卡，并隔离原生探测崩溃，让启动器显示可处理的错误而不是直接消失。
- **CPU 回退**：入门级硬件（如 Intel N100 系列）也能跑。
- 一键 **测速** 校准 ETA 估算，让等待时间贴合你这台机器。

🎙️ 本地离线音视频转文字 (STT)

- **SenseVoice Small**（默认，242 MB）— 中英日韩粤多语种，非自回归架构，CPU 上速度快。
- **Parakeet TDT 0.6B**（可选，661 MB）— 英文专用，NVIDIA 出品，WER 约 1.7%。
- **Silero VAD 精准分段** — 时间戳准确、零重叠、零卡顿。
- **实时滚屏字幕** SSE 流式输出；支持导出 TXT、SRT、VTT。

🛡️ 智能首次下载，断点续传

- **HTTP Range / resume 内置** — 一次 WiFi 闪断不会让 2.2 GB 的 GGUF 下载从零开始。
- **3 次重试，指数退避** — pip 依赖与模型文件都有同样的兜底逻辑。
- **国内友好镜像**：pip 按阿里云 → 清华 → PyPI 依次回退，某个镜像不可达或拒绝本机 IP 时自动换源；GGUF 运行模型走 ModelScope；STT 模型走 hf-

mirror.cn。

📱 局域网共享，手机平板可访问

- 首次启动询问是否绑定 `0.0.0.0`，允许同一 WiFi 下其他设备访问。
- 手机浏览器输入电脑的局域网 IP 即可使用，跟本地网页一样流畅。

🚀 快速上手

最简流程

1. 把下载的 zip 解压到任意文件夹（桌面或随便哪里都行）。
2. 在解压后的顶层目录里 **双击** **`Start_启动.bat`**。
3. 首次启动会：
    - 安装 Python 依赖（约 700 MB，根据网速 / CPU 不同需要 8-15 分钟）
    - 缺少 GGUF 模型时自动下载（约 2.2 GB，2-15 分钟；可按 Ctrl+C 取消，下次启动会续传）
    - 自动在默认浏览器打开 `http://127.0.0.1:5050`
4. 首次完成后，**之后每次启动都是离线**，5 秒左右即可打开。

进阶

如果想跳过顶层启动器，可以直接运行 `bashi-privacy-app\run_portable.bat`，效果完全一样。

🌐 网络行为说明

隐私承诺：本程序不主动联网、不上传音频、不收集使用数据。**程序永远不会自动检查任何在线资源。** 所有联网行为均在下文有完整说明，且需要用户主动触发 — 无后台轮询、无统计分析、无静默更新探测。

仅首次启动（一次性，约 700 MB + 约 2.2 GB）：

- pip 依赖来自 `https://mirrors.aliyun.com/pypi/simple/`，失败时依次回退清华、PyPI（海外用户仅用 PyPI）
- GGUF 模型来自 `https://modelscope.cn/models/gtree592/bashi-qwen3-tts-1.7b-customvoice-gguf-runtime`

- STT 模型（用户主动点击下载）来自 `https://hf-mirror.com/csukuangfj/...`（备用 HuggingFace）

运行时网络行为：零。文字转语音、音视频转文字、音频导出全部本机完成。

可选 CUDA 加速（NVIDIA 用户，仅用户主动触发）：检测到 NVIDIA 显卡时，UI 会显示”升级到 CUDA 加速”横幅。点击后从 `https://modelscope.cn/models/gtree592/bashi-qwen3-tts-cuda-runtime` 下载一次性 ~595 MB CUDA 运行时附加包。完全可选 — 不下载也能用 Vulkan 路线。不点就不会下载任何东西。

手动更新检查（仅用户主动触发，永不自动）：UI 右下角”查看最新版本”按钮在新浏览器标签页打开发布页。也可以随时手动访问：

- GitHub Releases: https://github.com/gtree965/bashi-voice-factory-privacy/releases
- files.fm 镜像: https://files.fm/u/juvstxmrez

离网 / 敏感场景部署（银行、医疗、政务、企业内网）：在联网机器上完成首次启动的所有下载后，整个 `bashi-voice-factory-privacy-v0.x.0/` 解压文件夹可以原样拷贝到离网/物理隔离机器上运行，之后所有功能均无需任何网络访问。程序没有任何遥测端点需要防火墙拦截，没有自动更新守护进程，也没有回调 URL。可用 Wireshark / 资源监视器在正常使用过程中监测出站流量验证 — 应当全程静默。

💻 硬件要求

下表是**作者实测**的数据。首次运行后点右侧面板的”测速”按钮，程序会根据你这台机器实际表现重新校准 ETA 估算 — 不用照搬此表。

硬件	自动选择的后端	“你好。” 探测（25 字）	1,000 字合成预计	状态
AMD RX 9060 XT (16 GB)	GGUF + Vulkan	约 3 秒	几分钟	✓ 2026-05 实测
AMD RX 590 (8 GB)	GGUF + Vulkan	3-5 秒	约 5-10 分钟	✓ 实测
Intel N305 笔记本 + UHD 集显	GGUF + Vulkan / DirectML	53 秒	25-46 分钟	✓ 2026-05-25 实测
Intel N100 迷你主机 + UHD	GGUF + Vulkan /	126 秒	58 分钟 - 1 小时 49 分	✓ 2026-05-25 实测

硬件	自动选择的后端	“你好。” 探测（25 字）	1,000 字合成预计	状态
集显	DirectML			
NVIDIA RTX 5070（12 GB，云电脑，Blackwell）	GGUF + Vulkan/CUDA + DirectML	约 1 秒	41 秒 - 1 分 17 秒	✓ 2026-06-12（Vulkan）& 2026-07-24（海马云 HMv Cloud PC）实测：全冷启动合成确认；可选 CUDA 附加包已实证可用（LLM 28 层跑 CUDA0、解码 DirectML fp16)。CUDA-vs-Vulkan A/B 仍欢迎反馈。
其它 NVIDIA RTX / GTX（桌面）	GGUF + Vulkan + DirectML · 可选 CUDA 附加包	待社区报数	待社区报数	等待桌面 tester 通过 GitHub Issues 报数
Apple Silicon / Intel Arc	—	暂未实测	暂未实测	尚未验证

> ⚠️ **入门级 CPU（Intel N100 / N305 一类）仅适合短句试用 — 单次生成建议不超过 200 字。** 这类机器跑 5,000 字长文需要 2-9 小时。如果想做长文音频（讲座、有声书），请用独立显卡硬件（AMD RX 500/600/9000 系、NVIDIA RTX 类）。

> ℹ️ **NVIDIA 用户（桌面 RTX / GTX)**：默认路线是 GGUF + Vulkan（NVIDIA 驱动支持 Vulkan)。想要原生 CUDA 加速，点击应用内一键升级横幅（v0.1.1+）— 会下载 ~595 MB CUDA 运行时附加包，无需手动配置权重。**需要 NVIDIA 驱动 ≥ 545.x 才能使用 CUDA 12.4 运行时。** Vulkan 路径在 RTX 5070（云电脑）实测：**25 字探测 1 秒**，桌面 NVIDIA 用户的默认

> Vulkan 体验已经非常顺畅。CUDA 附加包 A/B 数字 + 其它 RTX/GTX 卡的报数欢迎通过 GitHub Issues 反馈。云端数据中心 NVIDIA 显卡（A10/A100/T4）在 TCC 模式下需要额外设置 — 见下方常见问题排查。

🧩 硬件兼容性矩阵

除了上表里具体测过的几台机器，下表说明主流 Windows 硬件在自动后端选择下走的加速路径：

硬件类别	实际加速路径	状态
NVIDIA RTX 30 / 40 / 50（桌面）	默认 GGUF + Vulkan · 可选 CUDA 应用内升级	✅ RTX 5070（Blackwell）海马云云电脑已验证——2026-06-12 Vulkan 探测约 1 秒；2026-07-24 全冷启动合成 + 可选 CUDA 附加包实证可用（LLM 28 层跑 CUDA0、解码 DirectML fp16、驱动 610.47）。CUDA-vs-Vulkan A/B 数字 + 其它卡报数仍欢迎通过 Issues 反馈。CUDA add-on 需驱动 ≥ 545.x。
NVIDIA GTX 10 / 16（桌面）	默认 GGUF + Vulkan · 可选 CUDA 应用内升级	✅ 同一流程，同样驱动要求；报数欢迎通过 Issues。
NVIDIA 数据中心卡（A10 / A100 / T4）	需要手动配置	⚠️ TCC 模式 + 云端老驱动 = 需要手动 workaround。见常见问题排查。
AMD RX 500 / 600 / 7000 / 9000（独立显卡）	GGUF + Vulkan + DirectML	✅ 实测（RX 590、RX 9060 XT）
Intel Arc A 系列（A380 / A580 / A750 / A770）	GGUF + Vulkan + DirectML	✅ 理论可用、未实测
Intel 集显（UHD / Iris Xe / Arc iGPU）	GGUF + Vulkan + DirectML	✅ 实测（Intel N305 UHD）

硬件类别	实际加速路径	状态
AMD APU 集显（Vega 7/8、RDNA 2/3）	GGUF + Vulkan + DirectML	✅ 理论可用、未实测
纯 CPU（无可用 GPU / 无驱动）	GGUF + CPU（ggml-cpu 自动 SIMD）	✅ 可用；仅在无 GPU 时自动选择
NPU（Snapdragon X / Lunar Lake / Ryzen AI 300）	—	❌ 暂未利用
ARM64 Windows（Snapdragon X Copilot+ PC）	—	❌ 暂不支持（走 x64 仿真较慢）

⚙️ 已知限制（路线图）

- **NVIDIA CUDA 为可选项，不内置。** NVIDIA 显卡默认走 Vulkan 路线；原生 CUDA 加速需点击应用内一键升级（~595 MB 附加包，v0.1.1+）。这样主下载包对 AMD / Intel / CPU 大多数用户保持精简。需 NVIDIA 驱动 ≥ 545.x。
- **弱集显可能比纯 CPU 还慢。** Intel N100 类硬件上，DirectML / Vulkan 的数据传输与调度开销可能盖过 GPU 收益。可在 cmd 窗口里 `set GGUF_LLM_USE_GPU=0 && set GGUF_ONNX_PROVIDER=CPU` 后再启动来 A/B 测试，欢迎反馈实测数字。
- **NPU 加速尚未利用**（Snapdragon X Elite、Intel Lunar Lake AI Boost、AMD Ryzen AI 300）。v0.2+ 调研中。
- **ARM64 Windows 暂未原生支持**（Surface Pro 11、Galaxy Book4 Edge 等）。x64 仿真能跑但速度受限。原生 ARM64 构建是 v0.3+ 候选。

🐛 常见问题排查

现象	可能原因 / 处理
双击 `Start_启动.bat` 报 “Array index expression is missing”	旧版 zip。请从 GitHub Releases 或 files.fm 镜像重新下载最 版；该 BOM 问题在 v0.1.0 正式版已修复
pip 安装途中 WiFi 闪断后中止	自动重试 3 次（5/30/120 秒退避）。3 次都失败时，修好网纟 重新运行启动器即可（pip 会跳过已装好的包）

现象	可能原因 / 处理
pip 安装失败并提示 “long path support”	按启动器提示，在管理员 PowerShell 中运行一次注册表命 然后重新启动
所有镜像都装不上，或镜像拒绝本机 IP（`403` / `denied by IP ACL`）	启动器已按阿里云 → 清华 → PyPI 自动换源。若要指定某一源，启动前设置 `BASHI_PIP_INDEX_URL`，例如 `set BASHI_PIP_INDEX_URL=https://pypi.tuna.tsinghua.edu.cn/si`
GGUF 下载中断	同样自动重试，且支持 HTTP Range 续传，重新运行启动器 `.part` 文件续传
启动报 “No usable backend was found”	查 `app.log`（带时间戳的结构化应用日志）和 `launch_log.txt` 动器步骤与原生 stderr）。常见原因：GGUF 运行 DLL 缺失、卡驱动过旧、可用内存 <8 GB。程序会打印中英双语提示告 体原因
STT 下载闪过 “镜像失败”	v0.1.0 正式版不会出现 — 是旧版残留的 ModelScope 路径，删除
云端 / 数据中心 NVIDIA 显卡（A10 / A100 / T4，国内云 Windows 镜像）：`access violation reading 0x0000000000000000` 或 `GGUF probe failed`	数据中心 NVIDIA 显卡通常运行在 **TCC 模式**（Vulkan/Direct 被屏蔽），且驱动版本较旧（~538.x）跟 CUDA 12.4 add-on 容。**桌面 RTX/GTX 显卡不受影响**（默认 WDDM 模式 + 现代 动）。v0.1.2 会隔离并报告原生启动探测崩溃，不再让启动器 退出，但 TCC 配置仍需手动 workaround：(1) 把 `vulkan_backend_spike\Qwen3-TTS-GGUF\qwen3_tts_gguf\inference\bin\ggml-vulkan.dll` 重命名 `.disabled`；(2) 编辑 `bashi-privacy-app\run_portable.ps1`，7 `Remove-Item Env:USE_GGUF_BACKEND` 行（约第 270 行）之后加 `$env:USE_GGUF_BACKEND = "1"` + `$env:GGUF_ONNX_PROVIDER = "CPU"`；(3) 命令行预装 CUDA add-on：`python download_cuda_runtime.py`。CUDA 12.4 需要 NVIDIA 驱动 ≥ 545.x。TCC 自动处理仍计划在后续补丁实现。

完整日志路径：`bashi-privacy-app\app.log` 和 `bashi-privacy-app\launch_log.txt`

📦 zip 包目录结构

```
bashi-voice-factory-privacy-v0.1.3/
├── Start_启动.bat                                      ← 双击这里
├── Start_CPU_only_仅CPU启动.bat                        ← 强制 CPU 模式（入门
集显 A/B 对比用）
```

```
├── README.md                                      ← 英文文档
├── README_CN.md                                   ← 本文件
├── LICENSE                                        ← MIT 许可证
├── VERSION                                        ← 发布版本号
├── 巴适声工厂隐私版使用手册
_Bashi_Voice_Factory_Privacy_Edition_User_Guide.pdf
│                                                   ← 中英双语帮助 PDF
├── bashi-privacy-app/                             ← 程序代码 + 嵌入式
Python
│   ├── run_portable.bat                           ← 等效启动器（直接路
径）
│   └── ...
└── vulkan_backend_spike/                          ← GGUF 运行时（首次
启动自动填充）
    └── Qwen3-TTS-GGUF/
```

📄 许可

第三方组件保留各自原始许可：

- **Qwen3-TTS-12Hz-1.7B-CustomVoice**：通义实验室（阿里巴巴）Apache 2.0
- **GGUF runtime**：基于 HaujetZhao/Qwen3-TTS-GGUF
- **SenseVoice Small / Silero VAD / Parakeet TDT**：详见各自模型卡

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欢迎通过 GitHub release 页面或邮件提交 issue、反馈或功能建议。